

CUSTOMER SHOPPING BEHAVIOR ANALYSIS

A Data Analytics Portfolio Project

End-to-End Analysis using Python, PostgreSQL and Power BI

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1. Executive Summary

This report presents an end-to-end data analytics project performed on a customer shopping behavior dataset comprising 3,900 transaction records. The primary goal of the project is to demonstrate a complete data analyst workflow - beginning with raw data ingestion, moving through data cleaning, data preprocessing and exploratory data analysis (EDA) in Python, followed by structured SQL analysis in PostgreSQL, and concluding with the design of a business-ready Power BI dashboard for data-driven decision making.

The analysis is organized so that every stage follows the same logical flow used by professional data analysts: a specific business question is identified, an analytical technique is chosen to answer it, the technique is implemented using the appropriate tool, the resulting output is examined, and finally the result is translated into a business insight that a non-technical stakeholder can act on. This structure - Business Question -> Analysis -> Technical Approach -> Result -> Insight -> Business Meaning - is applied consistently throughout Section 6 of this report.

Overall, the analysis shows measurable differences in revenue contribution across gender and age groups, a customer base that is heavily skewed toward loyal, repeat purchasers, and a subscription program whose uptake does not strongly correlate with purchase frequency or spend - all of which carry direct implications for marketing and retention strategy.

2. Objective of the Project

The objective of this project is to apply core data analyst skills to a realistic retail dataset and produce actionable insights that could inform business strategy. Specifically, the project aims to:

- Perform data cleaning and data preprocessing on raw transactional data using Python and Pandas to ensure data quality and consistency prior to analysis.
- Conduct exploratory data analysis (EDA) to understand the structure, distribution and completeness of the dataset before drawing conclusions from it.
- Apply data transformation and feature engineering techniques to derive analytically useful fields, such as customer age groups, that are not present in the raw data.
- Persist the cleaned dataset into a relational database (PostgreSQL) so that it can be queried using SQL in a structured, repeatable and auditable manner.
- Use SQL-based data aggregation, subqueries, CASE-based segmentation and window functions to answer ten distinct business questions relevant to a retail stakeholder.
- Design an interactive Power BI dashboard that surfaces key performance indicators (KPIs) and supports self-service, filter-driven exploration for business users.
- Translate every technical result into a clear business insight, in line with how a professional data analyst would communicate findings to non-technical decision-makers.

3. Tools & Technologies Used

Each tool used in this project plays a distinct, well-defined role in the overall analytics pipeline. Only the tools actually applied in this project are described below.

Stage	Tool / Technology
Data Cleaning & Feature Engineering	Python (Pandas, Jupyter Notebook)
Database	PostgreSQL (pgAdmin 4)
Database Connectivity	SQLAlchemy, psycopg2
Data Analysis	SQL (Aggregations, CTEs, Window Functions)
Visualization / Dashboard	Microsoft Power BI

Python and Pandas

Python, using the Pandas library, was used as the primary tool for data cleaning, data preprocessing and exploratory data analysis. Pandas provides efficient, tabular (DataFrame-based) operations that make it well suited for detecting missing values, standardizing column formats, and engineering new features from raw fields - all essential first steps before any meaningful analysis can be performed.

Jupyter Notebook

Jupyter Notebook was used as the development environment for the Python workflow. Its cell-by-cell execution model allowed each preprocessing step (inspection, cleaning, transformation) to be run, verified and documented incrementally, which is a standard practice for exploratory data analysis and reproducible data preparation.

PostgreSQL and pgAdmin 4

PostgreSQL was used as the relational database to store the cleaned dataset in a structured, queryable form. Storing data in a proper database, rather than repeatedly reading a flat CSV file, reflects how analytics is typically performed in a business environment, where data must be centrally accessible, consistent, and query-able by multiple stakeholders. pgAdmin 4 was used as the SQL client to write, execute and validate the ten analytical queries covered in Section 6.

SQLAlchemy and psycopg2

SQLAlchemy, together with the psycopg2 driver, was used as the connectivity layer between the Python environment and the PostgreSQL database, enabling the cleaned Pandas DataFrame to be written directly into a PostgreSQL table without manual export/import steps.

Power BI

Microsoft Power BI was used to build the final interactive dashboard. Power BI converts the underlying SQL analysis into a visual, KPI-driven interface that supports slicing and filtering by attributes such as subscription status, gender, category and shipping type - allowing business

stakeholders without SQL knowledge to independently explore the same insights derived in Section 6.

4. Dataset Description

The dataset used in this project is a customer shopping behavior dataset containing 3,900 rows and 18 original columns, with each row representing a single customer transaction. Understanding the structure and quality of the dataset before performing any analysis is a foundational step of exploratory data analysis (EDA), as it determines what cleaning and transformation steps are required.

Attribute	Description
Customer ID	Unique identifier for each customer
Age / Gender	Demographic details
Item Purchased / Category	Product and its category (Clothing, Footwear, Outerwear, Accessories)
Purchase Amount (USD)	Transaction value
Location, Size, Color, Season	Order attributes
Review Rating	Customer's product rating (2.5 - 5.0)
Subscription Status	Whether the customer is a subscriber
Shipping Type, Discount Applied, Promo Code Used	Fulfilment & promotion details
Previous Purchases	Count of prior purchases by the customer
Payment Method, Frequency of Purchases	Payment and repeat-purchase behavior

Table 1: Raw dataset attributes (18 columns, 3,900 rows)

Summary statistics obtained during exploratory data analysis show an average purchase amount of \$59.76, an average review rating of 3.75, an age range of 18-70 years, and coverage across 50 unique customer locations and 25 unique products spanning 4 product categories. These summary figures act as reference values that are cross-checked against the SQL analysis and dashboard results later in this report, forming part of the data validation process.

5. Methodology

The methodology follows the standard analytics lifecycle used in professional data analyst workflows: data import and inspection, data cleaning, data transformation and feature engineering, and finally loading the processed data into a database for structured querying.

5.1 Data Import & Inspection (Python)

The dataset was first loaded into a Pandas DataFrame and inspected using `df.info()` and `df.describe()`. This exploratory step is a core part of EDA, as it reveals the data types, value ranges, and completeness of each column before any cleaning decisions are made. The inspection revealed 3,900 records across 18 columns, with the Review Rating column containing 37 missing values while all other columns were complete - information that directly guided the cleaning strategy in the next step.

```
import pandas as pd
df = pd.read_csv('customer_shopping_behavior.csv')
df.info()
df.describe(include='all')
df.isnull().sum()
```

5.2 Data Cleaning

Data cleaning is the process of identifying and correcting missing, inconsistent or poorly formatted data so that downstream analysis is accurate and reliable. Here, the 37 missing Review Rating values were imputed using the median rating within each product Category, rather than a single global median. This technique preserves category-specific rating patterns (for example, Footwear ratings are not distorted by Accessories ratings) and is a more analytically sound approach than blanket imputation.

```
df['Review Rating'] = df.groupby('Category')['Review Rating']\
    .transform(lambda x: x.fillna(x.median()))
```

Column names were then standardized to lowercase with underscores, and 'Purchase Amount (USD)' was renamed to 'purchased_amount'. Standardizing naming conventions at this stage is a routine data preprocessing step that prevents case-sensitivity and syntax errors during the SQL analysis phase that follows.

```
df.columns = df.columns.str.lower().str.replace(' ', '_')
df = df.rename(columns={'purchase_amount_(usd)': 'purchased_amount'})
```

As part of data validation, the Discount Applied and Promo Code Used columns were checked and found to be identical for every single record, using a boolean equality check across the full dataset. Since a fully redundant column adds no analytical value and only increases the dimensionality of the dataset unnecessarily, Promo Code Used was dropped.

```
(df['discount_applied'] == df['promo_code_used']).all()    # -> True
df = df.drop('promo_code_used', axis=1)
```

5.3 Data Transformation & Feature Engineering

Feature engineering is the process of deriving new, analytically meaningful variables from existing raw data. Two features were engineered here to support customer segmentation and trend analysis in later stages:

- **age_group** - customers were divided into four equal-sized bins (Young Adult, Adult, Middle-aged, Senior) using quartile-based binning (`pd.qcut`) on the Age column. This is a common data transformation technique used to convert a continuous numeric variable into a categorical one suitable for group-wise comparison and dashboard filtering.
- **purchase_frequency_days** - the categorical Frequency of Purchases field (e.g. Weekly, Monthly) was mapped to an approximate numeric number of days between purchases, converting a qualitative label into a quantitative measure suitable for future trend analysis.

```
labels = ['Young Adult', 'Adult', 'Middle-aged', 'Senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels=labels)

frequency_mapping = {'Weekly':7, 'Fortnightly':14, 'Monthly':30,
                     'Quarterly':90, 'Annually':365}
df['purchase_frequency_days'] =
df['frequency_of_purchases'].map(frequency_mapping)
```

5.4 Loading Data into PostgreSQL

Once cleaned and enriched, the DataFrame (3,900 rows x 19 columns) was loaded into a PostgreSQL database using SQLAlchemy, creating a table named `customer` within the `customer_behavior` database. Moving the dataset from a flat file into a relational database is a deliberate design choice that reflects real-world analytics practice: it enables the use of SQL for repeatable, auditable and performant querying, and forms the technical foundation for the SQL analysis presented in Section 6.

```
from sqlalchemy import create_engine

engine = create_engine(
    'postgresql+psycopg2://postgres:<password>@localhost:5432/customer_behavior'
)
df.to_sql('customer', engine, if_exists='replace', index=False)
```

6. SQL Analysis & Results

This section presents ten SQL-based analyses performed in pgAdmin 4 against the `customer` table. Each analysis follows a consistent structure - Business Question, Analysis Objective, Technical Approach, Result, and Business Insight - so that the technical SQL work is always connected back to a decision-relevant conclusion, which is the core responsibility of a data analyst.

Query 1 - Business Question: What is the total revenue generated by Male vs. Female customers?

Analysis Objective: To compare total revenue contribution across genders, which is a basic but essential customer segmentation exercise used to understand which demographic group drives the majority of sales.

Technical Approach: SQL data aggregation using GROUP BY with the SUM() aggregate function, grouping all transactions by the gender column.

```
SELECT gender, SUM(purchased_amount) AS revenue
FROM customer
GROUP BY gender;
```

gender	revenue (\$)
Female	75,191
Male	157,890

Result Interpretation: Male customers contributed \$157,890 in total revenue compared to \$75,191 from female customers - more than double.

Business Insight: Since male customers are the dominant revenue segment, marketing budgets and personalized promotions could be weighted toward this group, while a separate investigation into why female customer revenue is comparatively lower (e.g. product range, pricing, targeting) may reveal a growth opportunity.

Query 2 - Business Question: Which customers used a discount but still spent more than the average purchase amount?

Analysis Objective: To identify customers who are not purely price-driven - i.e. those who used a discount but still spent above the overall average - which is useful for distinguishing genuine deal-seekers from high-value customers who simply also use discounts.

Technical Approach: A correlated filter using WHERE combined with a scalar subquery (SELECT AVG(purchased_amount) FROM customer) to dynamically compare each row against the dataset-wide average purchase amount.

```
SELECT customer_id, purchased_amount
FROM customer
WHERE discount_applied = 'Yes'
AND purchased_amount >= (SELECT AVG(purchased_amount) FROM customer);
```

Result: 839 customers matched this condition (average purchase amount = \$59.76). Sample output:

customer_id	purchased_amount (\$)
2	64
3	73
4	90
7	85
9	97
12	68
13	72

customer_id	purchased_amount (\$)
16	81
20	90
22	62

839 rows returned in total (first 10 shown).

Result Interpretation: A substantial group of 839 customers (roughly 22% of the customer base) used a discount while still spending above the average purchase amount.

Business Insight: Discount usage does not automatically indicate a low-value, price-sensitive customer. This segment represents an actionable insight for the business: these customers respond well to discounts without needing them to spend a meaningful amount, making them strong candidates for future promotional campaigns.

Query 3 - Business Question: Which are the top 5 products with the highest average review rating?

Analysis Objective: To identify the products with the strongest customer satisfaction, which supports inventory, merchandising and quality-assurance decisions.

Technical Approach: SQL data aggregation using GROUP BY with the AVG() function on review_rating, combined with ORDER BY ... DESC and LIMIT to rank and restrict the result to the top five products.

```
SELECT item_purchased, ROUND(AVG(review_rating::numeric), 2) AS "Average Product Rating"
FROM customer
GROUP BY item_purchased
ORDER BY AVG(review_rating) DESC
LIMIT 5;
```

item_purchased	Average Product Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Handbag	3.78

Result Interpretation: Gloves, Sandals, Boots, Hat and Handbag received the highest average customer ratings, all above 3.78 out of 5.

Business Insight: These products can be prioritized in marketing campaigns and homepage placement, since they are already validated by customer satisfaction data - a data-driven way to decide what to promote rather than relying on assumption.

Query 4 - Business Question: Compare the average purchase amount between Standard and Express shipping.

Analysis Objective: To determine whether the shipping method a customer selects is associated with a difference in how much they spend, which can inform shipping-related pricing and promotion strategy.

Technical Approach: SQL data aggregation using GROUP BY with AVG(), filtered using a WHERE ... IN clause to restrict the comparison to only Standard and Express shipping types.

```
SELECT shipping_type, ROUND(AVG(purchased_amount), 2)
FROM customer
WHERE shipping_type IN ('Standard', 'Express')
GROUP BY shipping_type;
```

shipping_type	avg_purchase_amount (\$)
Express	60.48
Standard	58.46

Result Interpretation: Express shipping customers spend marginally more on average (\$60.48) than Standard shipping customers (\$58.46).

Business Insight: The difference is small, indicating shipping method is not a strong predictor of order value on its own. Express shipping could still be positioned as a modest upsell opportunity at checkout, but it should not be treated as a primary lever for increasing average order value.

Query 5 - Business Question: Do subscribed customers spend more? Compare average spend and total revenue between subscribers and non-subscribers.

Analysis Objective: To evaluate whether the subscription program is associated with higher customer spend, which is directly relevant to assessing the return on investment of the subscription initiative.

Technical Approach: SQL data aggregation combining COUNT(), AVG() and SUM() in a single GROUP BY query, with results ordered using ORDER BY for readability.

```
SELECT subscription_status,
       COUNT(customer_id) AS total_customers,
       ROUND(AVG(purchased_amount), 2) AS avg_spend,
       ROUND(SUM(purchased_amount), 2) AS total_revenue
FROM customer
GROUP BY subscription_status
ORDER BY total_revenue, avg_spend DESC;
```

subscription_status	total_customers	avg_spend (\$)	total_revenue (\$)
Yes	1,053	59.49	62,645
No	2,847	59.87	170,436

Result Interpretation: Average spend is nearly identical between subscribers (\$59.49) and non-subscribers (\$59.87). Non-subscribers generate more total revenue only because they are a larger group (2,847 vs 1,053 customers).

Business Insight: The subscription program does not currently drive higher individual spend, which is an important and slightly counter-intuitive finding for business-driven decision making. This suggests the subscription's value proposition may need to be redesigned - for example, offering spend-based incentives - rather than assuming subscribers are naturally higher-value customers.

Query 6 - Business Question: Which 5 products have the highest percentage of purchases with discounts applied?

Analysis Objective: To identify which products are most frequently purchased using a discount, which helps evaluate whether certain products are being over-discounted relative to others.

Technical Approach: SQL data aggregation using a conditional CASE expression inside SUM() to count discounted purchases, divided by COUNT(*) and scaled to a percentage, grouped by item_purchased.

```
SELECT item_purchased,
       ROUND(100 * SUM(CASE WHEN discount_applied='Yes' THEN 1 ELSE 0
END)/COUNT(*), 2) AS discount_rate
FROM customer
GROUP BY item_purchased
ORDER BY discount_rate DESC
LIMIT 5;
```

item_purchased	discount_rate (%)
Hat	50
Coat	49
Sneakers	49
Sweater	48
Pants	47

Result Interpretation: Hats, Coats, Sneakers, Sweaters and Pants have the highest discount usage rates, all between 47% and 50% of their respective purchases.

Business Insight: Close to half of all purchases of these products involve a discount, which may be compressing profit margins on these specific items. This is an actionable insight for the pricing or merchandising team to review discount policy on these five products specifically, rather than applying blanket discount rules.

Query 7 - Business Question: Segment customers into New, Returning, and Loyal based on total previous purchases.

Analysis Objective: To perform customer segmentation based on purchase history, which is a standard technique used to group customers into meaningful behavioral categories for targeted marketing and retention strategy.

Technical Approach: A Common Table Expression (CTE) combined with a CASE statement to classify each customer into a segment based on their previous_purchases value, followed by a GROUP BY COUNT() to summarize segment sizes.

```
WITH customer_type AS (
  SELECT customer_id, previous_purchases,
    CASE
      WHEN previous_purchases = 1 THEN 'New'
      WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
      ELSE 'Loyal'
    END AS customer_segment
  FROM customer
)
SELECT customer_segment, COUNT(*) AS "Number of Customers"
FROM customer_type
GROUP BY customer_segment;
```

customer_segment	Number of Customers
New	83
Returning	701
Loyal	3,116

Result Interpretation: The vast majority of customers, 3,116 out of 3,900 (approximately 80%), fall into the 'Loyal' segment, having made more than 10 previous purchases. Only 83 customers are classified as 'New'.

Business Insight: The customer base is dominated by loyal, repeat purchasers, which indicates strong customer retention overall. This also means acquisition of genuinely new customers is comparatively low, suggesting the business may benefit from investing more in top-of-funnel acquisition strategies alongside its already-strong retention performance.

Query 8 - Business Question: What are the top 3 most purchased products within each category?

Analysis Objective: To identify the best-selling products within each product category, which supports category-level inventory planning and merchandising decisions rather than a single flat 'best-seller' list.

Technical Approach: A CTE combined with the window function ROW_NUMBER() OVER (PARTITION BY category ORDER BY COUNT(customer_id) DESC), which ranks products independently within each category partition - a technique that a simple GROUP BY cannot achieve on its own.

```
WITH item_counts AS (
  SELECT category, item_purchased, COUNT(customer_id) AS total_orders,
    ROW_NUMBER() OVER (PARTITION BY category ORDER BY COUNT(customer_id) DESC) AS
```

```

item_rank
FROM customer
GROUP BY category, item_purchased
)
SELECT item_rank, category, item_purchased, total_orders
FROM item_counts
WHERE item_rank <= 3;

```

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Belt	161
3	Accessories	Sunglasses	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

Outerwear has only two distinct products in the dataset (Jacket, Coat), so only two ranks are returned for that category.

Result Interpretation: Jewelry leads Accessories, Blouse and Pants tie for the lead in Clothing, Sandals leads Footwear, and Jacket leads Outerwear.

Business Insight: These category leaders are natural candidates for prioritized stock allocation and prominent placement within their respective category pages, since they already have proven demand within their category rather than across the whole catalog.

Query 9 - Business Question: Are repeat buyers (more than 5 previous purchases) also likely to subscribe?

Analysis Objective: To test whether purchase loyalty (measured by previous purchase count) translates into subscription sign-up, which helps assess whether the subscription program is effectively capturing the business's most engaged customers.

Technical Approach: A filtered SQL data aggregation using WHERE previous_purchases > 5 combined with GROUP BY and COUNT() on subscription_status.

```

SELECT subscription_status, COUNT(customer_id) AS repeat_buyers
FROM customer
WHERE previous_purchases > 5
GROUP BY subscription_status;

```

subscription_status	repeat_buyers
No	2,518
Yes	958

Result Interpretation: Among repeat buyers with more than 5 previous purchases, 2,518 (about 72%) are still not subscribed, while only 958 (about 28%) are subscribers.

Business Insight: This confirms the pattern already observed in Query 5: purchase loyalty and subscription status are not strongly linked. This is a clear, data-driven opportunity - the business could specifically target its large pool of loyal, non-subscribed repeat buyers with a subscription conversion campaign, since they are already engaged and simply have not converted.

Query 10 - Business Question: What is the revenue contribution of each age group?

Analysis Objective: To perform trend analysis across customer age groups and determine which demographic segment contributes the most to total revenue, supporting age-targeted marketing decisions.

Technical Approach: SQL data aggregation using GROUP BY with SUM() on the engineered age_group column (created during feature engineering in Section 5.3), ordered using ORDER BY ... DESC.

```
SELECT age_group, SUM(purchased_amount) AS total_revenue
FROM customer
GROUP BY age_group
ORDER BY total_revenue DESC;
```

age_group	total_revenue (\$)
Young Adult	62,143
Middle-aged	59,197
Adult	55,978
Senior	55,763

Result Interpretation: Revenue is fairly evenly distributed across all four age groups, with Young Adult customers contributing marginally the most at \$62,143, and Senior customers the least at \$55,763.

Business Insight: Because revenue does not skew heavily toward any single age group, broad, age-inclusive marketing strategies are likely to be more effective than narrowly targeted age-based campaigns, though the slight lead of the Young Adult segment could still justify a modest tilt in campaign spend toward that group.

7. Power BI Dashboard

Analysis Objective: To consolidate the SQL analysis performed in Section 6 into a single, interactive, business-facing view, allowing stakeholders without SQL knowledge to explore the same insights independently.

Technical Approach: The cleaned dataset was imported into Power BI, where KPI cards, bar charts and a donut chart were built and linked to slicers for subscription status, gender, category and shipping type, enabling filter-driven, self-service exploration of the data.

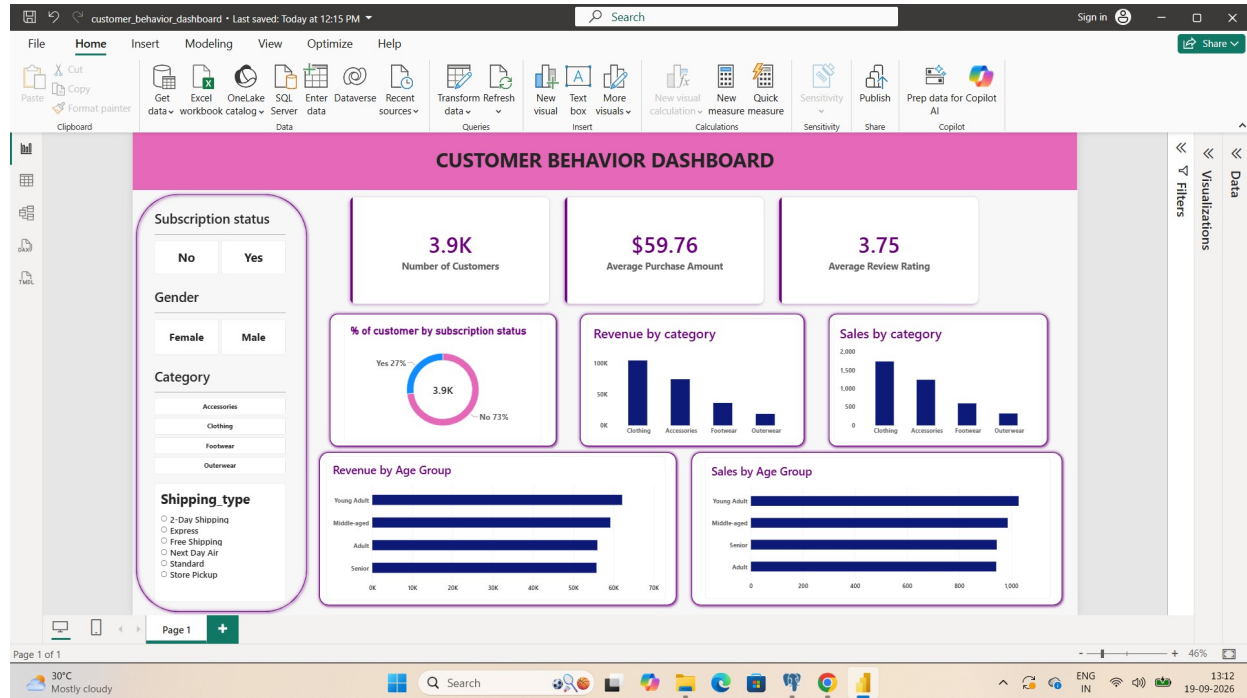


Figure 1: Customer Behavior Dashboard (Power BI) - KPI cards, subscription/gender/category filters, revenue & sales by category, and revenue & sales by age group.

Result Interpretation: The dashboard's KPI cards show 3.9K total customers, an average purchase amount of \$59.76, and an average review rating of 3.75, all of which are consistent with the dataset summary statistics in Section 4 and validate the underlying SQL analysis.

Additional dashboard-level observations:

- Only 27% of customers are subscribers, confirmed by the donut chart, matching the subscription split found in Query 5 and Query 9.
- Clothing leads both in revenue and sales volume by category, followed by Accessories, Footwear, and Outerwear - consistent with Query 8's category-level breakdown.
- Revenue and sales by age group are relatively balanced, with Young Adults slightly ahead, matching Query 10's results.

Business Insight: Because every figure on the dashboard is traceable back to the underlying SQL analysis, business stakeholders can trust the dashboard as a reliable, self-service decision-making tool - reducing their dependency on the data analyst for routine reporting and freeing analyst time for deeper, ad-hoc investigations.

8. Key Insights & Findings

The following actionable insights summarize the business-relevant conclusions drawn from the SQL analysis and dashboard in this report, and are intended to directly support data-driven decision making:

- Male customers generate roughly 2.1x the revenue of female customers (\$157,890 vs \$75,191), making gender a meaningful factor in revenue segmentation.
- 839 customers used a discount and still spent above the average purchase amount, suggesting discount usage does not necessarily indicate price-sensitive, low-spending customers.
- Gloves, Sandals, Boots, Hat and Handbag received the highest average review ratings, all above 3.78/5, making them strong candidates for promotional prioritization.
- Express shipping customers spend marginally more per order (\$60.48) than Standard shipping customers (\$58.46), though the difference is not large enough to be a primary revenue lever.
- Subscription status has little impact on individual spend - subscribers and non-subscribers have nearly identical average spend (~\$59-60); non-subscribers simply outnumber subscribers roughly 3:1.
- Hats, Coats and Sneakers have the highest discount usage rates (~48-50% of purchases), warranting a review of discount policy on these specific items.
- 80% of the customer base falls into the 'Loyal' segment (more than 10 previous purchases), indicating strong repeat-purchase behavior and customer retention.
- Even repeat buyers (>5 previous purchases) are mostly non-subscribers (72%), showing subscription uptake and purchase loyalty are not strongly linked - a clear conversion opportunity.
- Jewelry, Blouse/Pants and Sandals are the best-selling products in their respective categories, useful for category-level inventory planning.
- Young Adults are the highest revenue-contributing age group, though revenue is fairly evenly distributed across all four age groups.

9. Conclusion

This project demonstrates a complete, real-world data analyst workflow - from raw CSV data to a business-ready dashboard - applying data cleaning, data preprocessing, exploratory data analysis,

data transformation, SQL-based data aggregation, customer segmentation and trend analysis at each stage. Python and Pandas were used to clean and enrich the dataset; PostgreSQL and SQL enabled structured, repeatable querying to answer ten distinct business questions; and Power BI translated these findings into an interactive, KPI-driven dashboard suitable for non-technical stakeholders.

Throughout the report, every technical result has been deliberately connected back to a business insight, following the Business Question -> Analysis -> Technical Approach -> Result -> Insight -> Business Meaning structure used by professional data analysts. The analysis shows that while subscription status and shipping type have only a marginal effect on individual spend, gender and age group are stronger differentiators of revenue, and the customer base overall skews heavily toward loyal, repeat buyers who have not yet converted into subscribers. These insights could inform targeted marketing - such as converting the large pool of loyal, non-subscribed repeat buyers - and inventory decisions that prioritize high-rating, high-demand products such as Gloves, Jewelry and Blouses, ultimately supporting more data-driven decision making across the business.